

DSAA3073: Theories in Data Science

Week 2 Validation Sheet: PAC Learning Framework & VC Dimension

Total Points: 130

Instructions:

- Write your answers clearly and legibly.
- Show all working for full credit.
- State any assumptions you make.
- For proofs, justify each step rigorously.
- This is a practice validation sheet. AI tools and references are allowed.
- Use this to assess your understanding before the actual test.

Part A: Multiple Choice (40 points)

Each question is worth 5 points. Select the single best answer.

Question 1. [5 marks] In the PAC learning framework, what does the parameter δ represent?

- A) The maximum allowed generalization error.
- B) The probability that the learning algorithm fails to produce a hypothesis with error $\leq \epsilon$.
- C) The size of the training sample.
- D) The VC dimension of the hypothesis class.

Question 2. [5 marks] Consider two hypothesis classes: $\mathcal{H}_1$ with $|\mathcal{H}_1| = 100$ and $\mathcal{H}_2$ with $|\mathcal{H}_2| = 10000$. In the realizable setting with $\epsilon = 0.1$ and $\delta = 0.05$, approximately how much more data does $\mathcal{H}_2$ require compared to $\mathcal{H}_1$?

- A) 100 times more
- B) About twice as much
- C) About 10 times more
- D) About 40-50 additional samples (not multiplicative)

Question 3. [5 marks] A set $S = \{x_1, x_2, x_3\}$ is said to be “shattered” by hypothesis class $\mathcal{H}$ if:

- A) There exists at least one hypothesis $h \in \mathcal{H}$ that correctly classifies all points in S .
- B) Every hypothesis $h \in \mathcal{H}$ makes at least one error on S .
- C) For every possible labeling of the 3 points (all $2^3 = 8$ labelings), there exists a hypothesis $h \in \mathcal{H}$ that realizes that labeling.
- D) The growth function satisfies $\Pi_{\mathcal{H}}(3) = 3$.

Question 4. [5 marks] Which statement about the growth function is TRUE?

- A) $\Pi_{\mathcal{H}}(m)$ can exceed 2^m for some hypothesis classes.
- B) If $\Pi_{\mathcal{H}}(m) = 2^m$ for all m , then $\text{VCdim}(\mathcal{H}) = \infty$.
- C) $\Pi_{\mathcal{H}}(m)$ depends on the data distribution $\mathcal{D}$.
- D) For finite $\mathcal{H}$, we always have $\Pi_{\mathcal{H}}(m) = |\mathcal{H}|$.

Question 5. [5 marks] For the hypothesis class of axis-aligned rectangles in $\mathbb{R}^2$, what is the VC dimension?

- A) 2
- B) 3
- C) 4
- D) 5

Question 6. [5 marks] What is the key difference between the consistent learner and Empirical Risk Minimization (ERM)?

- A) Consistent learner works in agnostic setting; ERM works in realizable setting.
- B) Consistent learner returns any hypothesis with zero training error; ERM returns the hypothesis with minimum training error.
- C) Consistent learner minimizes generalization error; ERM minimizes training error.
- D) They are identical algorithms.

Question 7. [5 marks] Sauer's lemma shows that if $\text{VCdim}(\mathcal{H}) = d$, then:

- A) $\Pi_{\mathcal{H}}(m) = d$ for all m .
- B) $\Pi_{\mathcal{H}}(m) \leq \left(e \frac{m}{d}\right)^d$, which is polynomial in m .
- C) $|\mathcal{H}| \leq 2^d$.
- D) $\Pi_{\mathcal{H}}(m) = 2^m$ for $m \leq d$.

Question 8. [5 marks] Which hypothesis class has infinite VC dimension?

- A) Linear classifiers in $\mathbb{R}^{\{100\}}$
- B) Decision trees with maximum depth 10
- C) $\mathcal{H} = \{\text{sign}(\sin(\omega x)) : \omega \in \mathbb{R}\}$ on $\mathbb{R}$
- D) Polynomials of degree at most 5 in $\mathbb{R}$

Part B: Mixed Format Questions (90 points)

Question 9. [12 marks] Consider a finite hypothesis class $\mathcal{H}$ in the agnostic PAC learning setting with $|\mathcal{H}| = 1000$.

- (a) [4 marks] What is the goal of learning in the agnostic setting? How does it differ from the realizable setting?
- (b) [4 marks] Derive the sample complexity bound for the agnostic setting.
- (c) [4 marks] Compute the required sample size for $\varepsilon = 0.1$ and $\delta = 0.05$.

Question 10. [15 marks] Consider the hypothesis class $\mathcal{H} = \{h_{a,b} : h(x) = \mathbb{1}_{a \leq x \leq b}, a \leq b\}$ of closed intervals on $\mathbb{R}$.

- (a) [5 marks] Compute $\Pi_{\mathcal{H}}(4)$ for 4 points $x_1 < x_2 < x_3 < x_4$ by enumerating which dichotomies are realizable.
- (b) [5 marks] Prove that $\text{VCdim}(\mathcal{H}) = 2$ using a shattering argument.
- (c) [5 marks] Verify that Sauer's lemma holds for this class: check that $\Pi_{\mathcal{H}}(4) \leq \sum_{i=0}^2 \binom{4}{i}$.

Question 11. [13 marks] Prove or disprove the following statements about VC dimension:

- (a) [4 marks] “If $\text{VCdim}(\mathcal{H}_1) = d_1$ and $\text{VCdim}(\mathcal{H}_2) = d_2$, then $\text{VCdim}(\mathcal{H}_1 \cup \mathcal{H}_2) = \max(d_1, d_2)$.”
- (b) [4 marks] “A hypothesis class with 10 parameters always has $\text{VCdim} \leq 10$.”
- (c) [5 marks] “If a hypothesis class has infinite cardinality, it must have infinite VC dimension.”

Question 12. [15 marks] Consider linear classifiers in $\mathbb{R}^3$: $\mathcal{H} = \{\text{sign}(w_1x_1 + w_2x_2 + w_3x_3 + b) : w_i, b \in \mathbb{R}\}$.

- (a) [7 marks] Prove that $\text{VCdim}(\mathcal{H}) \geq 4$ by constructing a set of 4 points that can be shattered.
- (b) [8 marks] State Radon’s theorem for $\mathbb{R}^3$ and explain how it implies that no set of 5 points in $\mathbb{R}^3$ can be shattered by linear classifiers.

Question 13. [15 marks] Compare sample complexity across different settings.

- (a) [5 marks] For a hypothesis class with $|\mathcal{H}| = 2^{\{20\}}$, compute the sample complexity in the realizable setting with $\varepsilon = 0.05$ and $\delta = 0.01$.
- (b) [5 marks] For the same hypothesis class in the agnostic setting with the same parameters, compute the sample complexity.
- (c) [5 marks] Now suppose $\mathcal{H}$ has $\text{VCdim}(\mathcal{H}) = 20$ but infinite cardinality. Using the VC bound $m = O\left(\frac{d}{\varepsilon^2}\right)$ (ignoring log factors for simplicity), approximately how many samples are needed? Compare with parts (a) and (b).

Question 14. [10 marks] Explain the significance of the union bound in PAC learning theory.

- (a) [5 marks] In the proof of the finite hypothesis class bound, we use the union bound over $|\mathcal{H}|$ hypotheses. Explain why we need the union bound and what would go wrong without it.
- (b) [5 marks] In the transition from finite to infinite hypothesis classes, how does Sauer's lemma help us apply the union bound? What replaces $|\mathcal{H}|$ in the bound?

— End of Validation Sheet —